

MAT 1272

Sections 13.1 and 13.4: Scatterplots, Correlation, and Regression

Based on Sections 13.1 and 13.4 of the textbook. Section 13.1.7 is omitted.

Learning Objectives

Goals for Today

By the end of this lesson, students should be able to:

- identify independent and dependent variables;
- interpret scatterplots;
- recognize positive, negative, and no correlation;
- interpret the correlation coefficient r ;
- understand the regression equation $\hat{y} = a + bx$;
- interpret slope and y-intercept;
- understand why the regression line is a “best-fit” line;
- use the TI-84 Plus to find a , b , and r .

1. Scatter Diagrams

Scatter Diagram

A **scatter diagram**, or **scatterplot**, is a graph of paired observations.

Each point has the form

$$(x, y).$$

The horizontal axis represents the independent variable.

The vertical axis represents the dependent variable.

Example: Hours Studied and Exam Score

Hours Studied	Exam Score
1	58
2	64
3	70
4	73
5	81

Here,

x = hours studied

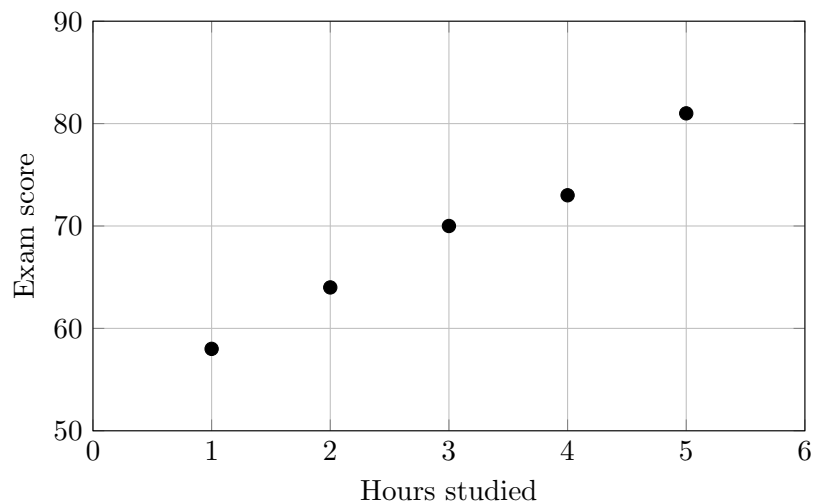
and

y = exam score.

The ordered pairs are

$(1, 58), (2, 64), (3, 70), (4, 73), (5, 81)$.

Scatterplot: Hours Studied vs. Exam Score



Visual Interpretation

The points show an upward trend.
As hours studied increase, exam scores tend to increase.
This suggests a **positive relationship**.

2. Independent and Dependent Variables

Independent Variable

The **independent variable** is the explanatory variable.
It is usually denoted by

x .

Dependent Variable

The **dependent variable** is the response variable.
It is usually denoted by

$$y.$$

Classroom Language

The independent variable helps explain or predict the dependent variable.
For example, hours studied may help predict exam score.

3. Correlation

Correlation

Correlation describes the strength and direction of a linear relationship between two variables.

Types of Correlation

Positive correlation: as x increases, y tends to increase.

Negative correlation: as x increases, y tends to decrease.

No correlation: there is no clear linear pattern.

4. Correlation Coefficient

Correlation Coefficient

The correlation coefficient is denoted by

$$r.$$

It satisfies

$$-1 \leq r \leq 1.$$

The closer r is to 1 or -1 , the stronger the linear relationship.

Interpreting r

Value of r	Interpretation
1	Perfect positive correlation
0.9	Strong positive correlation
0.7	Moderate positive correlation
0	No linear correlation
-0.7	Moderate negative correlation
-0.9	Strong negative correlation
-1	Perfect negative correlation

Examples of Correlation

If

$$r = 0.94,$$

then there is a strong positive linear relationship.

If

$$r = -0.88,$$

then there is a strong negative linear relationship.

If

$$r = 0.02,$$

then there is little or no linear relationship.

5. Least-Squares Regression Line

Least-Squares Regression Line

The **least-squares regression line** is the line that best fits the scatterplot.

It is the line that minimizes the overall prediction errors.

The equation is

$$\hat{y} = a + bx.$$

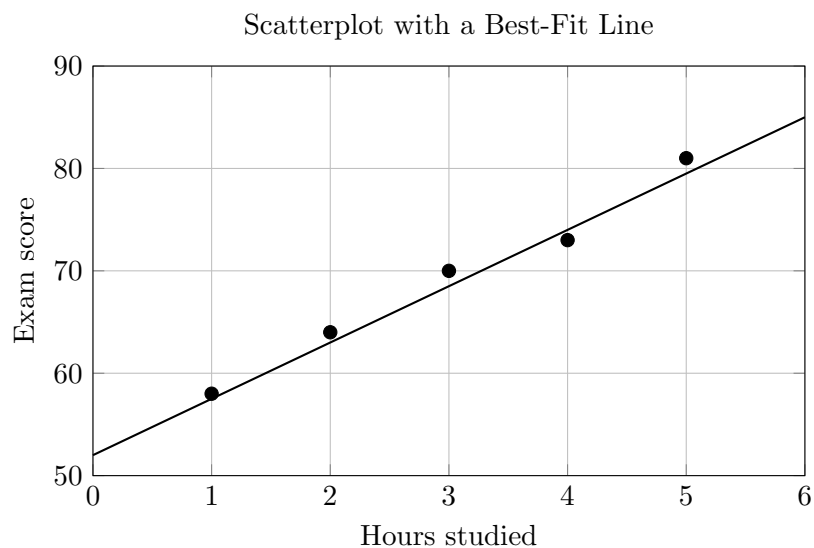
Meaning of the Symbols

In the equation

$$\hat{y} = a + bx,$$

- $\hat{y}$ is the predicted value of y ;
- a is the y-intercept;
- b is the slope;
- x is the independent variable.

6. Visual Meaning of the Regression Line



Best-Fit Line

A good regression line:

- follows the overall trend of the scatterplot;
- does not need to pass through every point;
- usually has some points above it and some points below it;
- is used to make predictions.

7. Prediction Error and Distance from the Line

Prediction Error

For each point, the regression line gives a predicted value

$$\hat{y}.$$

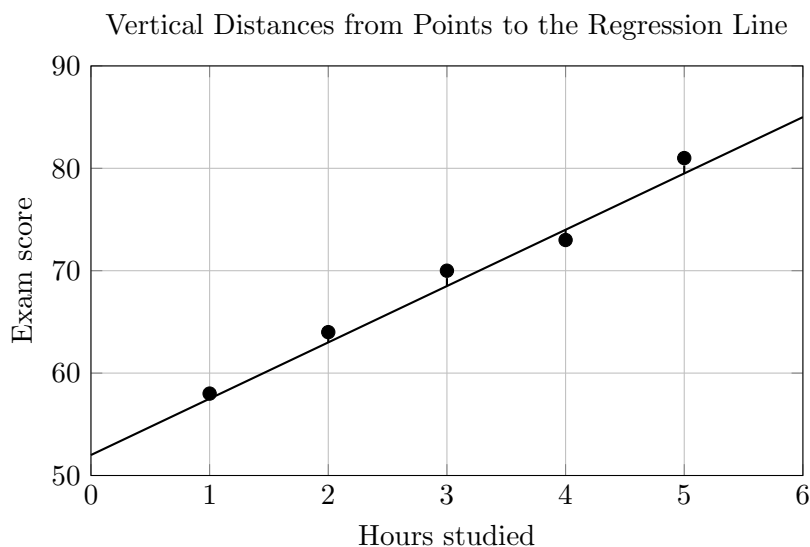
The actual value is

$$y.$$

The prediction error is

$$y - \hat{y}.$$

This is the vertical distance from the point to the regression line.



Least-Squares Idea

The least-squares line is chosen so that the squared prediction errors are as small as possible overall.

That is why it is called the **least-squares** regression line.

8. Formula for the Regression Coefficients

Formula for the Slope

The slope of the least-squares regression line is

$$b = \frac{SS_{xy}}{SS_{xx}},$$

where

$$SS_{xy} = \sum (x - \bar{x})(y - \bar{y})$$

and

$$SS_{xx} = \sum (x - \bar{x})^2.$$

For this course, we will use the TI-84 Plus calculator to compute b .

Formula for the Y-Intercept

Once the slope b has been found, the y-intercept is

$$a = \bar{y} - b\bar{x}.$$

For this course, we will use the TI-84 Plus calculator to compute a .

9. Slope

Slope

The slope tells us how much the predicted value of y changes when x increases by one unit.

Interpreting Slope

Suppose the regression equation is

$$\hat{y} = 52 + 4x.$$

The slope is

$$b = 4.$$

Interpretation:

For every additional hour studied, the predicted exam score increases by 4 points.

10. Y-Intercept

Y-Intercept

The y-intercept is the predicted value of y when

$$x = 0.$$

Interpreting the Y-Intercept

Suppose

$$\hat{y} = 52 + 4x.$$

When

$$x = 0,$$

we get

$$\hat{y} = 52 + 4(0) = 52.$$

Interpretation:

A student who studies 0 hours has a predicted exam score of 52.

11. Prediction

Prediction Example

Suppose

$$\hat{y} = 52 + 4x.$$

Predict the exam score for a student who studies 8 hours.

$$\hat{y} = 52 + 4(8)$$

$$\hat{y} = 52 + 32 = 84.$$

Therefore, the predicted exam score is

84.

Warning About Prediction

A regression line should only be used for prediction when the value of x is within the range of the observed data.

Using the regression line far outside the observed range is called extrapolation and can be misleading.

12. Group Activity

Work in Groups

A professor records the following data.

Hours Studied	Exam Score
1	60
2	65
3	70
4	74
5	80

Answer the following:

1. Identify the independent variable.
2. Identify the dependent variable.
3. Sketch a scatterplot.
4. Is the relationship positive or negative?
5. Would you expect r to be positive or negative?
6. Is r closer to 0, 0.5, or 1?
7. Use the TI-84 Plus to find the regression equation.
8. Interpret the slope in context.
9. Use the regression equation to predict the exam score for 6 hours of study.

13. TI-84 Plus Calculator Guide

Step 1: Enter the Data

Press:

STAT

Choose:

EDIT

Enter:

- x-values in L1;
- y-values in L2.

Step 2: Turn Diagnostics On

This step allows the calculator to show the value of r .

Press:

2nd

Then press:

0

This opens the Catalog.

Scroll down to:

DiagnosticOn

Press ENTER.

Press ENTER again.

The calculator should display:

Done

Step 3: Compute the Regression Equation

Press:

STAT

Move to:

CALC

Choose:

LinReg(ax+b)

Press ENTER.

The calculator displays:

- a = y-intercept;
- b = slope;
- r = correlation coefficient;
- r^2 = coefficient of determination.

Write the regression equation as

$$\hat{y} = a + bx.$$

Summary

Key Takeaways

- A scatterplot displays paired data.
- The independent variable is x .
- The dependent variable is y .
- Correlation describes strength and direction.
- The correlation coefficient satisfies
$$-1 \leq r \leq 1.$$
- The regression equation is
$$\hat{y} = a + bx.$$
- a is the y-intercept.
- b is the slope.
- The least-squares line minimizes the squared prediction errors.
- The TI-84 Plus gives a , b , and r .